Experimental Evaluation of Waveguide Group Index in a Mach-Zehnder Interferometer

jfoptic
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***Index Terms* — Mach-Zehnder interferometer, free spectra range, group index, silicon photonics.**

I. INTRODUCTION

THE Mach-Zehnder interferometer (MZI) is a fundamental photonic structure widely used for wavelength filtering, sensing, and dispersion characterization [1-3]. In this work, we designed [4] and fabricated MZIs with varying arm length differences to investigate free spectral range (FSR) behavior. By measuring the transmission spectra [5], the corresponding FSR values were extracted for each device. Based on these results, the group index of the integrated waveguide was calculated and compared with theoretical expectations. This study demonstrates a practical approach for characterizing waveguide dispersion properties using simple interferometric structures.

II. THEORY

Figure 1 illustrates a MZI made of two Y-branch and two arms of waveguides with different lengths (L_1 and L_2). When an input light with electric field E_i enters the first Y-branch, it is equally divided into two arms, denoted as

$$E_{i1} = E_{i2} = E_i/\sqrt{2}, \quad (1)$$

each carrying half of the incident intensity. As the light propagates, the two arms accumulate different phase shifts due to their unequal optical path lengths. Assuming negligible absorption and scattering losses, the recombined fields at the second Y-branch can be written as

$$E_{o1} = \frac{E_i}{\sqrt{2}} \mathrm{e}^{i(\omega t - \beta L_1)}, \quad (2)$$

$$E_{o2} = \frac{E_i}{\sqrt{2}} \mathrm{e}^{i(\omega t - \beta L_2)}, \quad (3)$$

where $\beta = \frac{2\pi n}{\lambda}$ is the propagation constant. n is the refractive index and λ is the light wavelength.

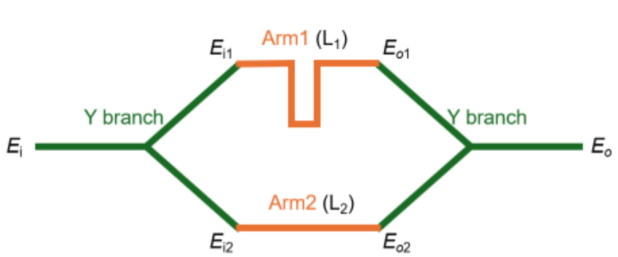


Fig. 1. Schematic of a Mach-Zehnder Interferometer.

When these two fields interfere at the output coupler, the resulting electric field is the superposition of both, leading to an output intensity that depends on the cosine of the phase difference between the two arms:

$$I_0 = {E_0}^2 = \frac{{E_i}^2}{4}(1 + \cos(\beta \cdot \Delta L)), \quad (4)$$

$$\Delta L = L_1 - L_2. \quad (5)$$

The free spectral range (FSR) of an MZI is defined as the wavelength spacing between adjacent transmission peaks. It can be expressed as:

$$FSR = \frac{\lambda^2}{\Delta L \cdot n_g}, \quad (6)$$

$$n_g = n - \lambda \frac{dn}{d\lambda}, \quad (7)$$

where n_g is the group index of the waveguide.

By measuring the FSR for devices with different ΔL, the group index can be extracted from the slope of the FSR versus ΔL relation. This provides a practical method for characterizing the dispersion properties of integrated photonic waveguides.

III. MODELLING AND SIMULATION

As the fundamental building block of the MZIs, we employed a strip waveguide with a height of 220 nm and a width of 500 nm, designed to operate at a wavelength of 1550 nm under TE polarization. The Si waveguide is fully embedded in SiO_2 claddings. The refractive index of Si and SiO_2 were taken from Palik and applied in the simulations. Figure 2 shows the quasi-TE mode supported by the waveguide, simulated using the Lumerical MODE solver.

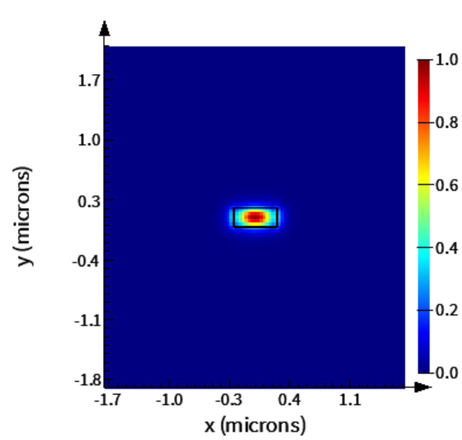


Fig. 2. Simulated *E* field distribution of the quasi-TE mode at 1550 nm supported by the employed waveguide.

The wavelength-dependent mode effective index and group index have also been extracted from the simulation, as shown in Fig. 3. The Taylor series expansion of the simulated effective index gives us a compact model to describe the waveguide properties:

$$n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2, \quad (8)$$

where we set λ_0 as 1550 nm, and get $n_1 = 2.45$, $n_2 = -1.13$, and $n_3 = -0.04$ from the results in Fig. 3.

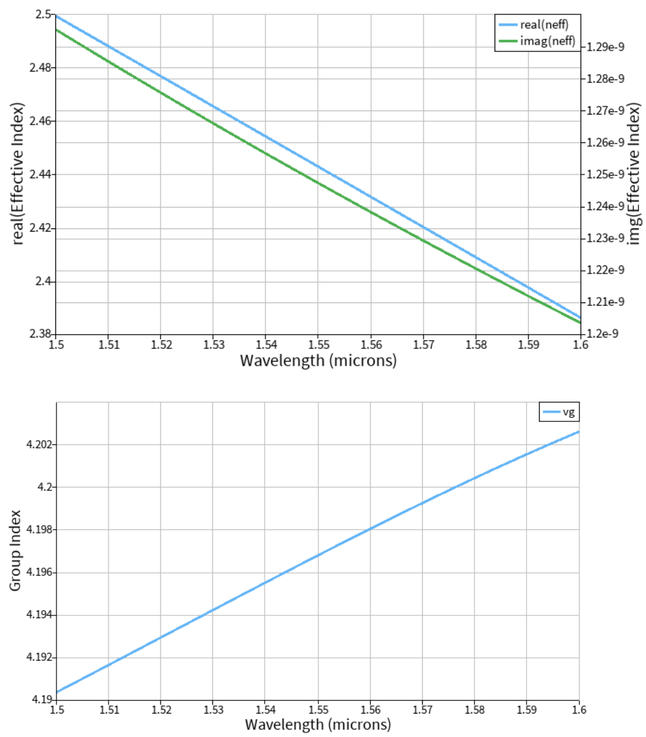


Fig. 3. Simulated mode (Top) effective index and (Bottom) group index as a function of wavelength for the quasi-TE mode in the employed waveguide.

Based on the simulated group index of ~4.20 at 1550 nm wavelength and Eq. 6, we can now estimate the FSR for MZI devices with different ΔL.

Six MZI devices with ΔL of 100, 200, 300, 400, 500, and 600 μm are studied in this work, and their estimated FSR are concluded in Table 1 below. Ideally, measuring these MZI devices will give us the same group index result. However, fabrication variation is inevitable, and we will further discuss the expected variation via corner analysis in Section V.

Device#	ΔL (μm)	FSR (nm)
1	100	5.72
2	200	2.86
3	300	1.91
4	400	1.43
5	500	1.14
6	600	0.95

Table. 1. Estimated FSR for different MZI devices.

IV. Design and Layout

Figure 4 shows the GDS Layout of all six MZI devices we will fabricate and study. The light signal input is always through the most top grating coupler, and the output signals towards the two other grating couplers will both be collected and analyzed.

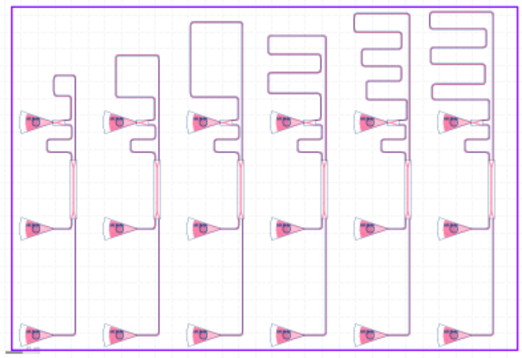
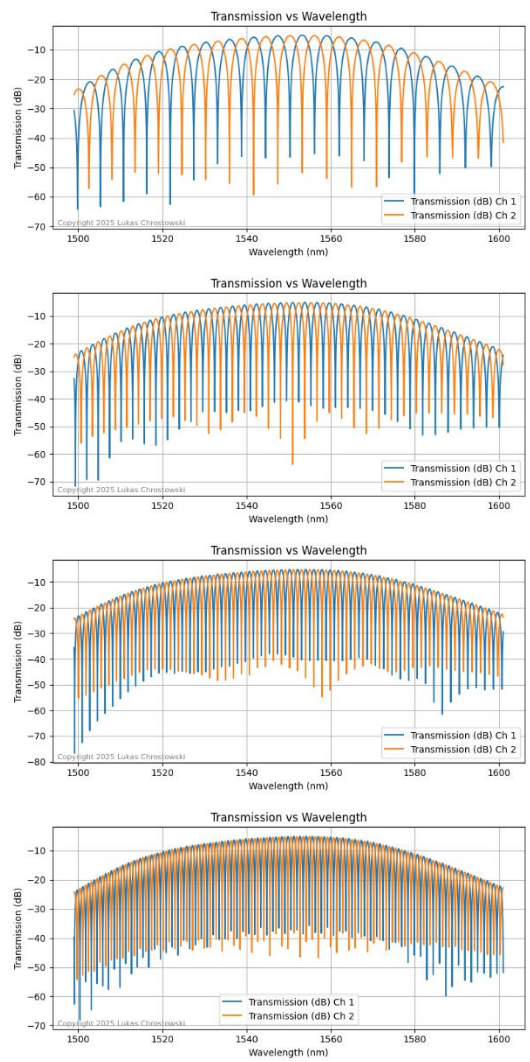


Fig. 4. GDS Layout of all six MZI devices.

Figure 5 illustrates the simulated output signals using Lumerical INTERCONNECT solver, as a guidance for experimental tests.



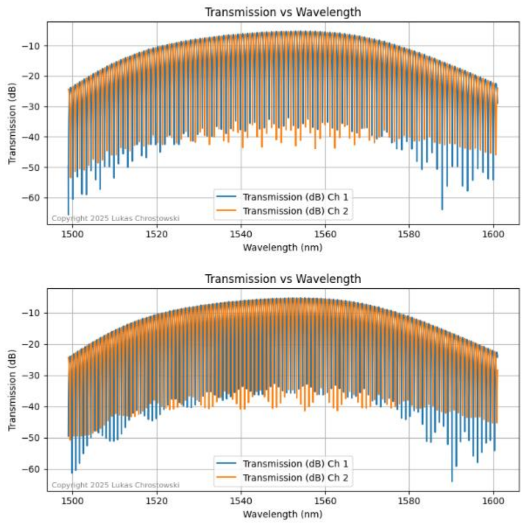


Fig. 5. Simulated signals at both output grating couplers for six MZI devices 1 to 6 from top to bottom.

V. Fabrication Method and Variation

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (http://www.appliednt.com/nanosoi; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 µm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H2SO4:H2O2) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent.

The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4” handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality.

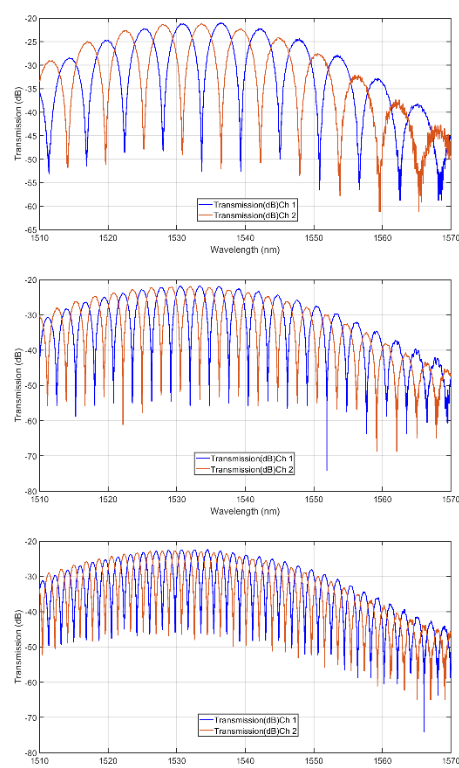
A 2.2 µm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300ºC. Reflectrometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

According to the previous experience on this fabrication process, we expect the wafer thickness varies between 215.3 and 223.1 nm and the waveguide width varies between 470 and 510 nm. Applying these values, we did a corner analysis of the expected variation in waveguide group index through Lumerical simulation. The expected group index varies between 4.158 and 4.237 for our TE mode under study. We will see it experimentally whether the measured group index falls in this range.

VI. Measurement Results

To characterize the devices, a custom-built automated test setup [4, 6] with automated control software written in Python was used [7]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [8]. A polarization maintaining fibre array was used to couple light in/out of the chip [9].

The results for all 6 MZI devices are shown in Fig. 6 below. It is qualitatively consistent with the simulation in Fig. 5 ---- as ΔL increases from device 1 to 6, the FSR in the transmission spectra decreases. In addition, the MZI responses (sharp dips) are modulated by a much broader peak (baseline/background) centered at ~1535 nm. This mainly comes from the wavelength dependent grating coupler efficiency and the measurement setup is also non-uniform at different wavelengths. Later in the fittings, we will manually extract such a broad baseline/background peak. The extraction of such an MZI-unrelated baseline/background can also be done by measuring a reference device with only two grating couplers and a straight waveguide.



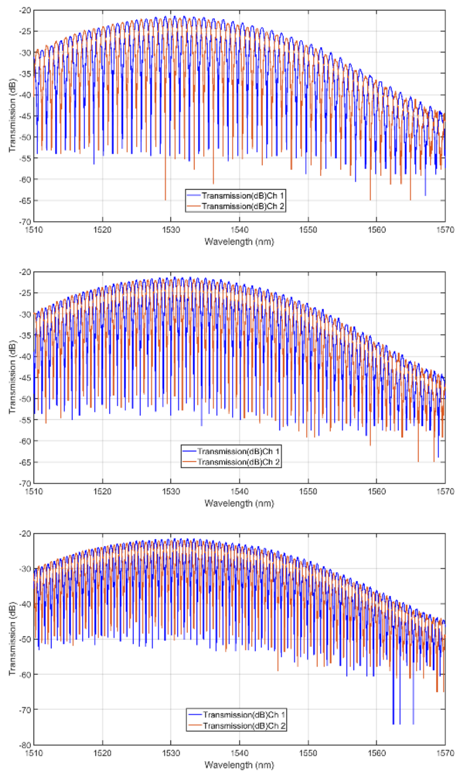
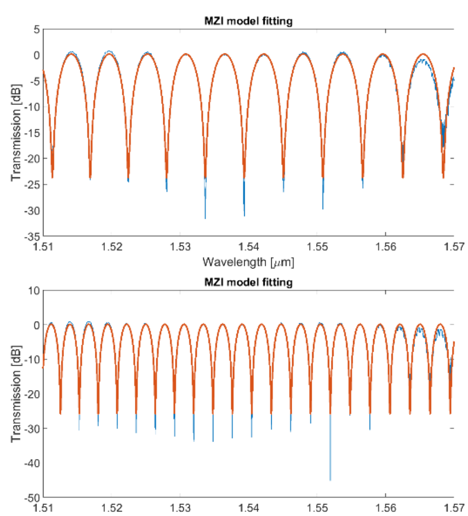


Fig. 6. Experimentally measured signals at both output grating couplers for six MZI devices 1 to 6 from top to bottom.

VI. Fittings and Data Analysis

To quantitively understand the real device performance, get the waveguide group index, and evaluate the fabrication variation, we fit the spectra in Fig. 6. Note that we first find the local minimum for each dip, then perform autocorrection based on these dips to manually remove the baseline, and finally apply the MZI model fitting (Eq. 6) on the modified spectra.

The fitting results are concluded in Fig. 7 below. The blue curves are the modified spectra after the extraction of baseline. The orange curves are the fitted results. They all show >90% good fitting accuracy.



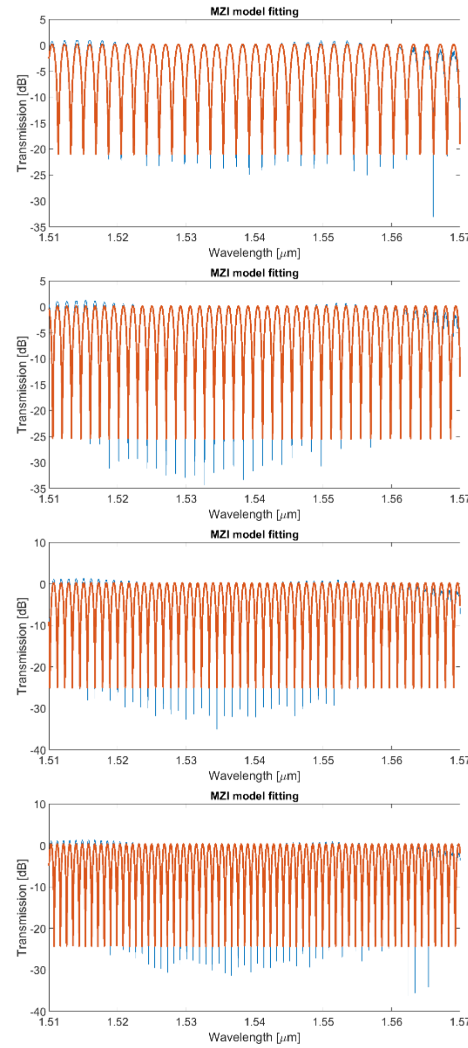


Fig. 7. MZI model fittings of the modified experimental spectra after baseline/background extraction for six MZI devices 1 to 6 from top to bottom.

Before looking into the extracted waveguide group index from each device, it is worth discussing the potential factors that influence the experimental results. First, the fabrication variation could happen both on single device (deviation from the designed geometry parameters) as well as among different devices (each device has different geometry deviation values). Second, the real ΔL for extracting group index value in Eq. 6 could be different from the designed values. It is reasonable to believe that such waveguide length difference deviation $\Delta\Delta L$ does not increase with ΔL (more like a fabrication error constant). Therefore, a device with larger ΔL should give us more accurate group index.

Table 2 below provides all the extracted group indexes from the experimental spectra fittings in Fig. 7. Interestingly, from device#1 with the smallest ΔL, we get a group index quite different from other devices, which is even outside of the variation range we expected in corner analysis in Section V. We believe this is due to the waveguide length difference deviation $\Delta\Delta L$. An SEM measurement of the real ΔL in device#1 can be helpful. For now, we will simply focus on devices#2 to #6.

Device#	ΔL (μm)	n_g
1	100	4.154
2	200	4.172
3	300	4.178
4	400	4.182
5	500	4.184
6	600	4.185

Table. 2. Experimentally extracted group indexes from different MZI devices.

A group index of 4.180 by averaging the results from devices#2 to #6 is concluded. It is well sitting in the expected variation range according to our corner analysis. Compared to the simulation value for an ideal waveguide, 4.177, the experimental discrepancy is only 0.07%, showing the good fabrication accuracy and the robustness of this MZI method in extracting the waveguide properties.

VI. Conclusion

In conclusion, we designed and fabricated MZIs with varying arm length differences to investigate free spectral range (FSR) behavior. By measuring the transmission spectra, the corresponding FSR values were extracted for each device. Based on these results, the group index of the integrated waveguide was calculated to be 4.180. Compared with theoretical expectations, 4.177, the experimental discrepancy is only 0.07%, showing the good fabrication accuracy and the robustness of this MZI method in extracting the waveguide properties. This study demonstrates a practical approach for characterizing waveguide dispersion properties using simple interferometric structures.

References

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